

Chapter 71 — Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

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PHYSIOLOGY OF PENILE ERECTION

Historical Contexts for Our Understanding of Erectile Physiology

- Chapter epigraph (Leonardo da Vinci): the penis **acts independently of its owner's will**, rising freely even during sleep — the first hint of an involuntary, autonomic event.
- **c. 2000 BCE, Egyptian papyrus** — oldest recorded description of erectile dysfunction (ED); two types: **natural** (inability to accomplish the sex act) and **supernatural** (evil charms and spells).
- **Hippocrates** — ED among wealthy Scythians, ascribed to **excessive horseback riding**.
- **Aristotle** — three nerve branches carry spirit and energy to the penis; erection produced by **influx of air**. Accepted until da Vinci.
- **Leonardo da Vinci** — erection derived from **penile blood flow**, from the blood-filled erect penis of hanged men; writings unknown until the early 20th century.
- **Ambroise Paré (1585)** — accurate anatomy (concentric coats of nerves, veins and arteries; two ligaments = corpora cavernosa; a urinary tract; four muscles) and the concept of **blood rushing into the member with lust**.
- **Dionis** — stressed **retention** of blood; attributed it to muscular compression of proximal penile veins.
- **Hunter (1787)** — **venous spasm** prevents outflow during erection.
- **1970s** — **xenon washout and cavernosography** during audiovisual sexual stimulation; technical limits gave conflicting results:

CONFLICTING 1970S–1980S HAEMODYNAMIC HYPOTHESES

Shirai et al. (1978)	Wagner (1981)
Venous outflow increased during erection, compensated by markedly greater arterial inflow	Venous drainage decreased during erection

- **Resolution: Wagner was right** — Doppler work (Lue) confirmed decreased outflow via venous coaptation.
- **1980s** — **Doppler ultrasound (Lue et al., 1984–1986)**: erection is driven by **increased arterial inflow (arterial dilation) plus decreased venous outflow (venous coaptation)**.
- **1990s** — **nitric oxide (NO)** and its synthesising enzymes, **endothelial and neuronal nitric oxide synthase (eNOS, nNOS)**, identified as key regulators (**Rajfer et al., 1992**).
- **1996** — **anti-erectile activity of phosphodiesterase type 5 (PDE5)** identified (**Boolell et al.**) → **1998** — **oral PDE5 inhibitors (PDE5Is)** for ED (**Goldstein et al.**).
- Later determinants: **gap junctions, potassium and calcium channels, RhoA/Rho-kinase**; and the role of **tissue elasticity and fibrosis** in erection and detumescence.

Concept — *Theory moved from pneumatic to vascular to veno-occlusive to molecular, and each step relocated the therapeutic target, ending at the cyclic guanosine monophosphate breakdown enzyme that PDE5Is exploit.*

Functional Anatomy of the Penis

- **Three cylinders:** paired **dorsal corpora cavernosa** (pubic rami to tip) and a **ventral corpus spongiosum** encircling the urethra, beneath loose subcutaneous tissue and skin. All contain **spongy vascular tissue able to hold large blood volumes**.
- **Flaccid length and girth are set by corporal blood content**, which is set by the **degree of contraction of trabecular smooth muscle**.
- Length and girth vary widely; their relevance to satisfaction is **subjective and variable**.
- **Natural curvature varies:** downward curvature during erection in up to one in seven men, and about one-quarter have an **erect angle below horizontal**.

Corpora Cavernosa, Corpus Spongiosum, and Glans Penis

- Corpora cavernosa: paired expansile cylinders within the **tunica albuginea**. **Crura** arise separately from the undersurface of the **puboischial rami**, merge under the pubic arch and remain attached up to the glans.
- **Humans: intercorporeal connections present** (incomplete septum) — versus **dogs: septum completely separates** the corpora.
- Each corpus = interconnected **vascular sinusoids, larger centrally and smaller peripherally**, separated by **smooth muscle trabeculae** surrounded by elastic fibres, collagen and loose areolar tissue.
- **Spongiosum and glans: same structure, larger sinusoids**.
- **Terminal cavernous nerves and helicine arteries** are intimately associated with the corporal smooth muscle.
- **Flaccid:** slow diffusion central → peripheral sinusoids; **blood gases venous**. **Erect:** rapid arterial entry to all sinusoids **raises oxygen tension and pH**.

Tunica Albuginea

- **Bilayered fibrous sheath** giving flexibility, rigidity and tissue strength (Fig. 71.1).
- Principal constituent **fibrillar collagen — mostly type I, also type III; elastic fibres** form an irregular latticed network interwoven with collagen (Fig. 71.2) and permit **expansion and recoil**.

LAYERS OF THE CORPORAL TUNICA ALBUGINEA

Outer layer	Inner layer
Longitudinal bundles, glans → proximal crura	Circular bundles; support and contain cavernous tissue
Thicker dorsally ; accounts for the dorsal-ventral thickness difference; absent in the ventral groove housing the spongiosum	Gives rise to intracavernous pillars — struts augmenting the septum; insert into inferior pubic rami; absent between 5 and 7 o'clock

- **Oblique fibres** connect the two layers.
- **Ventral tunica thinner** (mean approximately 0.8 mm) than **dorsal** (approximately 2.2 mm). Thin tunica plus **absent pillars** → **ventral area of relative weakness**, relevant to **penile fracture and implant surgery**.
- **Veno-occlusive role:** **emissary veins run between inner and outer layers** for a short distance, piercing the outer bundles obliquely; the expanding sinuses compress them early in erection → obstruction of outflow → tumescence.
- **Cavernous artery and dorsal-artery branches are spared** this compression: they run within the corpora inside a **periarterial soft-tissue sheath**.
- **Spongiosal tunica lacks an outer layer and intracorporeal struts** → **low-pressure system; it does not extend into the glans**.

EXTERNAL LIGAMENTOUS SUPPORT OF THE CORPORA

	Fundiform ligament	Suspensory ligament
Origin	Colles fascia	Buck fascia
Relation	Lateral, superficial, not adherent to tunica	Divided into fundiform, suspensory, dense vertical and arcuate components
Function	—	Attaches tunica to pubis ; supports the mobile portion

- **Fig. 71.2** — Hart stain micrograph (×100) of the interwoven elastic and collagen network.

Concept — *The bilayer is a valve: the inner circular layer contains, the outer longitudinal layer traps the emissary veins, so wherever the outer layer or pillars are missing (ventral groove, spongiosum, glans) pressure cannot be held and the tissue is weaker.*

Arteries

- **Primary source: paired internal pudendal arteries** (branches of internal iliac; Fig. 71.3A).
- Sequence: **internal iliac → internal pudendal → (perineal branch) → common penile artery → cavernous, dorsal and bulbourethral branches → distal vascular ring near the glans.**

BRANCHES OF THE COMMON PENILE ARTERY

Branch	Supplies	Key features
Cavernous	Main supply of corpus cavernosum	Enters hilum where crura converge; gives helicine arteries to trabeculae and sinusoids — contracted and tortuous flaccid, dilated and straight erect ; open directly into sinusoids without capillaries (Fig. 71.3B)
Dorsal	Glans engorgement	In many men, perforating branches pierce the tunica to join cavernous arteries in the distal penis
Bulbourethral	Penile bulb and corpus spongiosum	Bulbourethral and urethral arteries lie outside the spongiosal tunica , lateral and dorsal

- **Accessory arteries in about one-third:** from external iliac, obturator, vesical and femoral arteries; may be the **dominant or only** cavernosal supply.
- Pattern: **mixed pudendal plus accessory in most (~70%)**; purely pudendal or purely accessory in about 15% each.
- **Preserving accessory pudendal arteries → faster recovery of sexual function after radical prostatectomy.**

Veins

- Sequence: **venules from peripheral sinusoids → trabeculae between tunica and peripheral sinusoids → subtunical venous plexus → emissary veins through the inner tunica** (Fig. 71.4A).
- **Wide individual variability** of venous drainage (Fig. 71.4B, C).

THE THREE PENILE VENOUS DRAINAGE SYSTEMS

System	Drains	Course and destination
Deep dorsal vein (occasionally paired)	Glans and distal two-thirds of corpora (pendulous penis)	Arises near coronal sulcus; runs cephalad, posterior to symphysis → periprostatic (Santorini) plexus . Tributaries: circumflex veins (lateral), periurethral veins (ventral), channels accompanying dorsal arteries; periarterial veins join dorsal vein or Santorini plexus proximally
Cavernous and crural veins	Proximal (infrapubic) penis	Formed by proximal emissary veins; join periurethral veins from the bulb → internal pudendal veins

System	Drains	Course and destination
Superficial dorsal vein (single or paired)	Skin and subcutaneous tissue	Subcutaneous veins unite at the base → saphenous veins ; occasionally drains part of the corpora

- **Fig. 71.4A** — corrosion cast: blue injected into corpus cavernosum, yellow into deep dorsal vein, skin and tunica digested with potassium hydroxide, showing an emissary vein arising from the subtunical plexus.

Concept — *Both arterial and venous anatomy are highly variable between men, which is why the functional consequences of pelvic surgery on the same nominal structures differ from patient to patient.*

Hemodynamics and Mechanism of Erection and Detumescence

- **Erection = sinusoidal relaxation + arterial dilation + venous compression** Seven phases of arterial flow and intracavernous pressure described in animal experiments (Fig. 71.5).

Corpora Cavernosa

- **Cavernous smooth muscle and arteriolar/arterial smooth muscle** are the key effectors. **Flaccid: tonically contracted**, admitting only a small arterial inflow.
- Sequence: **sexual stimulation** → **neurotransmitter release from cavernous nerve terminals** → **smooth muscle relaxation** → **arteriolar and arterial dilation** → **sinusoidal pooling compresses subtunical plexus (latent)** → **tunica expands; emissary veins compressed between inner circular and outer longitudinal layers (tumescence)** → **maximal occlusion, intracavernous pressure approximately 100 mm Hg (full erection)** → **reflex ischiocavernosus contraction, pressure several hundred mm Hg (rigid erection)** → **pressure above systolic reverses flow in the cavernous arteries.**

FIG. 71.5 — THE SEVEN PHASES OF ERECTION AND DETUMESCENCE

Phase	Arterial flow	Intracavernous pressure
0 Flaccid	Low	Low
1 Latent (filling)	Peaks sharply	Barely changes
2 Tumescence	Falls as pressure rises	Rises
3 Full erection	Low, above flaccid	Plateau (~100 mm Hg)
4 Rigid erection	Near zero (may reverse)	Suprasystolic — pudendal nerve drives ischiocavernosus
5 Initial detumescence	—	Transient rise
6 Slow detumescence	—	Slow fall
7 Fast detumescence	—	Rapid fall to baseline

- Cavernous nerve drives phases 1–3; **the pudendal (somatic) nerve drives phase 4.**
- **Trap: arterial inflow is highest in the latent phase, not at full rigidity.**
- **Angle of erection** set by size and attachments — crura to puboischial rami; suspensory and fundiform ligaments to pubis. **Large penis or loose suspensory ligament** → **points downward even at full rigidity.**
- **Detumescence — three overlapping phases:**
 - **Initial** — transient pressure rise as trabecular smooth muscle contracts against still-closed emissary veins.
 - **Slow** — slow pressure fall as emissary veins gradually reopen.
 - **Rapid** — rapid fall as venous outflow capacity returns to baseline.

Corpus Spongiosum and Glans Penis

- Arterial flow increases, but the **thin spongiosal tunica and absent glans tunica allow minimal venous occlusion**, so pressure stays low (Table 71.1).

- **Full erection:** partial compression of deep dorsal and circumflex veins between **Buck fascia and the engorged corpora** contributes to glans tumescence, but **spongiosum and glans act essentially as a large arteriovenous shunt**.
- **Rigid erection:** **ischiocavernosus and bulbocavernosus** forcefully compress spongiosum and penile veins → more engorgement and pressure.

TABLE 71.1 COMPARISON OF CORPUS SPONGIOSUM AND GLANS PENIS — REBUILT WITH CORPORA CAVERNOSA FOR CONTRAST

	Corpora cavernosa	Corpus spongiosum	Glans penis
Tunica albuginea	Thick bilayer with pillars	Thin, circular layer only	Absent
Main blood supply	Cavernous (helicine) arteries	Bulbar and spongiosal arteries	Dorsal artery
Venous occlusion during erection	Yes	No	No
Compression by skeletal muscle	Ischiocavernosus (rigid phase)	Yes (ischiocavernosus, bulbocavernosus)	No
Erect pressure	~100 mm Hg, suprasystolic when rigid	One-third to one-half of corporal	

- Source: Lue, 1986. Table tests **why only the corpora become rigid**: veno-occlusion exists only where the bilayered tunica exists.

KEY POINTS

- Erection depends on **increased arterial inflow and coaptation of venous structures**.
- The **structure of the corporal bodies** is essential to the erectile process.
- **Penile vascular anatomy varies substantially** between men, which may matter for preservation of erection after pelvic surgery.

Concept — *Rigidity is a pressure phenomenon produced by trapping, not by inflow: the corpora are a closed high-pressure chamber because their tunica occludes outflow, whereas the spongiosum and glans are open low-pressure shunts.*

Smooth Muscle Physiology

Cytosolic Calcium and the Calcium Sensitization Pathway

- **Corporal smooth muscle is contracted at rest.** Its myosin isoforms have **both phasic and tonic characteristics**.
- Contractions are **spontaneous or induced**. Spontaneous phasic contractions depend on an **endogenous pacemaker** — a **cytosolic Ca^{2+} oscillator** periodically releasing Ca^{2+} from the sarcoplasmic reticulum; modulated by neurotransmitters and hormones.
- **Two master controls of contraction:** **cytosolic free Ca^{2+}** and **Rho-kinase signalling**.

Cytosolic free calcium

- Contraction: **free Ca^{2+} binds calmodulin** → **conformational change, ↑ affinity for myosin light chain (MLC) kinase** → **MLC kinase activated** → **phosphorylates serine-19 of regulatory MLC_{20}** → **with adenosine triphosphate (ATP), actin activates myosin ATPase** → **cross-bridge cycling**; ATP hydrolysis supplies the energy (Fig. 71.7).

- Relaxation: ↓ sarcoplasmic Ca^{2+} → calmodulin dissociates from MLC kinase (inactive) → MLC_{20} dephosphorylated by myosin light chain phosphatase (MLCP).

Rho-kinase signalling pathway

- **MLCP inhibition enhances contraction** by preventing MLC_{20} inactivation. Rho/Rho-kinase is a constituent of the **calcium sensitisation pathway** in penile smooth muscle (Fig. 71.9).
- Sequence: **agonist** → RhoA dissociates from Rho-guanine dissociation inhibitor → Rho-kinase activated → phosphorylates the regulatory subunit of MLCP → phosphatase inhibited → contraction sustained.
- **Latch state**: tonic contraction also relies on dephosphorylated myosin remaining bound to actin in the **high-affinity state**; **calponin** binds actin and myosin, stabilising cross-bridges and slowing detachment. Critical for **basal tone**.
- Rat cavernous-nerve-injury model: Rho-kinase may drive **fibroblast** → **myofibroblast transformation** → **fibrosis**.

TWO ROUTES TO CONTRACTION ACTING ON ONE SUBSTRATE

Calcium-dependent route	Calcium sensitisation route
Switches MLC kinase on (Ca^{2+} -calmodulin)	Switches MLCP off (Rho-kinase)
Needs rising cytosolic Ca^{2+}	Sustains contraction without further Ca^{2+} rise
Phasic contraction	Tonic contraction, latch state

- **Fig. 71.7** — Contraction: noradrenaline from sympathetic endings, and endothelins, angiotensin II and prostaglandin $\text{F}_{2\alpha}$ from endothelium, act on smooth muscle receptors → ↑ intracellular Ca^{2+} , Rho-kinase activation, contraction. **Protein kinase C** regulates the Ca^{2+} -independent sustained phase of agonist-induced contraction.

Molecular Pathways That Directly Modulate Intracellular Free Calcium

VASOCONSTRICTOR AGONISTS AND RECEPTORS

Agonist	Receptor
Noradrenaline (norepinephrine)	α_1 -adrenergic
Endothelin-1	Endothelin-A
Angiotensin II	Angiotensin type 1 (AT1)
Prostaglandin $\text{F}_{2\alpha}$	FP
Thromboxane A_2 (TXA_2)	TP

- All are **Gq-coupled**.
- **Gq** → phospholipase C β → hydrolyses phosphatidylinositol 4,5-bisphosphate → inositol trisphosphate (IP_3) + 1,2-diacylglycerol → IP_3 receptor on smooth endoplasmic reticulum → Ca^{2+} release.
- IP_3 also **sensitises its receptor to Ca^{2+}** and assists Ca^{2+} -induced Ca^{2+} release.
- Extracellular Ca^{2+} entry through **voltage-dependent Ca^{2+} channels**.
- Noradrenaline, endothelin, vasopressin and angiotensin II activate Ca^{2+} -permeable nonselective cation channels.

Molecular Pathways That Indirectly Modulate Intracellular Free Calcium

- **Cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP)** are the relaxant second messengers. cAMP- and cGMP-dependent protein kinases phosphorylate proteins and channels, producing:
 - (1) potassium-channel opening and hyperpolarisation; (2) sequestration of Ca^{2+} by the endoplasmic reticulum; (3) inhibition of voltage-dependent Ca^{2+} channels → net fall in cytosolic Ca^{2+} → relaxation.

THE TWO RELAXANT SECOND-MESSENGER ARMS

	cGMP arm	cAMP arm
Principal activators	NO , including NO released from endothelium by acetylcholine (ACh); also CO , H_2S , natriuretic peptides	Prostaglandin E_1 (PGE_1) , vasoactive intestinal peptide (VIP), calcitonin gene-related peptide (CGRP), adenosine (A_2); forskolin
Cyclase	Guanylyl cyclase (soluble for NO)	Adenylyl cyclase (Gs-coupled)
Kinase	Protein kinase G (PKG-I)	Protein kinase A
Degraded by	PDE5 (principal)	PDE2, 3, 4
Drug exploiting it	PDE5Is	Intracavernosal PGE_1 ; papaverine (nonspecific PDE inhibitor) acts on both

Cyclic guanosine monophosphate–signalling pathway

- **Nitric oxide** — best-characterised cGMP activator. Small; **diffuses into target cells**; binds iron in heme or iron–sulfur complexes. **Physiological receptor: soluble guanylate cyclase (sGC)**, whose **heme moiety is essential** for NO-induced vasorelaxation.
- The NO–sGC–cGMP pathway mediates many **endothelium-dependent vasodilators**: histamine, oestrogens, insulin, corticotropin-releasing hormone, nitrovasodilators and ACh → **NO is the molecule principally responsible for initiating erection**.
- **NO synthase (NOS)**: $\text{L-arginine} + \text{O}_2 \rightarrow \text{L-citrulline} + \text{NO}$.

NITRIC OXIDE SYNTHASE ISOFORMS

Isoform	Preferential site	Role in erection
nNOS	Neurons and nerves	Initiates erection
eNOS	Endothelium	Sustains erection (with nNOS)
Inducible NOS (iNOS)	Almost all cell types	Present in corpus cavernosum; not the principal driver

- **eNOS activity and bioavailability** regulated by phosphorylation, regulatory-protein interaction, contractile pathways and **reactive oxygen species**.
- **NOS downregulation** accompanies ED in animal models of ageing, hypogonadism and diabetes.
- Humans: **single-nucleotide polymorphisms of nNOS and eNOS genes and specific microRNA patterns** associated with impaired erectile function and/or **differential response to erectogenic drugs**; microRNA changes also in hyperlipidaemic and aged rats.
- NO/cGMP also **inhibits antitumescence pathways**: RhoA/Rho-kinase, phospholipase C–derived IP_3 , and endothelin vasoconstriction.

THE GASEOUS SECOND MESSENGERS

	Nitric oxide	Carbon monoxide	Hydrogen sulfide
Source	NOS from L-arginine + O ₂	Heme oxygenase catabolism of heme (endothelial corporal tissue)	Cystathionine γ-lyase breakdown of L-cysteine (vascular tissue)
Mechanism	sGC → cGMP	Binds sGC heme → cGMP; also guanylate-cyclase-independent relaxation	K ⁺ channels; NOS-dependent and -independent; downstream via cGMP and RhoA-Rho-kinase-1
Other actions	Inhibits antitumescence pathways	Vessel tone, smooth muscle proliferation, platelet aggregation; possible neurotransmitter	Relaxes corpus strips; inhibits phenylephrine contraction; species differences
Key evidence	Principal erection initiator	Potentially important role	Intracavernosal sodium hydrosulfide erects diabetic and nerve-injured rats; men with ED have lower H₂S-synthesising enzymes

- **H₂S relaxation of human tissue is not blocked by NOS inhibition but is blocked by endothelium removal or sGC inhibition** — it acts through cGMP, not through NO synthesis.
- In rat models, sodium hydrosulfide raised cGMP, NO and heme oxygenase (diabetes), and improved smooth muscle-to-collagen ratio with lower RhoA-Rho-kinase-1 (nerve injury).
- **Natriuretic peptides** — atrial (ANP) and brain (BNP) → **natriuretic peptide receptor A (NPR-A = guanylyl cyclase A, GC-A)**; C-type (CNP) → **NPR-B (GC-B)**. **CNP is the most potent**; raises cGMP and cAMP in human corporal cells; **physiological role in erection undetermined**.
- **Guanylyl cyclase pathway** — **seven membrane-bound (particulate) isoforms, GC-A to GC-G, and one soluble isoform**. Particulate cyclases have **no known role in physiological erection**, although GC-B is expressed in corpus cavernosum and CNP relaxes it.
- **sGC is pivotal: it links NO (principal extracellular signal) to cGMP (principal intracellular signal)**. Heterodimer of **α and β subunits**, each with two isoforms (α₁, α₂, β₁, β₂) from separate genes.
- **Protein kinase G (PKG, cGMP-dependent kinase)** — principal cGMP receptor. PKG-I and PKG-II from separate genes; **smooth muscle expresses only PKG-I** (splice variants Iα, Iβ).
- cGMP/PKG-I relax smooth muscle by: **activating the plasma membrane Ca²⁺-ATPase pump; inhibiting IP₃ generation; inhibiting Rho-kinase; stimulating MLCP; phosphorylating heat shock proteins**.

KNOCKOUT CONTRAST — REDUNDANCY VERSUS BOTTLENECK

nNOS or eNOS knockout	PKG-I knockout
Erection often normal — compensation (eNOS higher in nNOS-knockout mice), alternative splicing of the disrupted gene, or unknown mechanisms	Erection fails — cavernous strips cannot be relaxed by cGMP-raising agents; poor reproduction, presumably ED

Cyclic adenosine monophosphate — cyclase, kinase and phosphodiesterases

- **Adenylyl cyclase**: ligands act on membrane receptors coupled to G proteins. All membrane adenylyl cyclases are stimulated by the **guanosine triphosphate (GTP)-bound Gα subunit**; **all except AC9 are stimulated by forskolin**.
- **Protein kinase A (PKA, cAMP-dependent kinase)** — principal cAMP receptor. In penis and pelvic ganglion it **phosphorylates nNOS at serine 1412**.
 - Structure: **tetramer R₂C₂**; cAMP binds regulatory subunits → **R₂(cAMP)₄ dimer + two free active catalytic subunits**. Multiple catalytic-subunit genes → isozyme diversity and broad specificity.

- **>100 substrates; >90% phosphorylated at serine**, the rest threonine; predominant motif **Arg-Arg-X-Ser**.
- **Three penile PKA substrates: phosphodiesterases, cAMP-responsive element-binding protein, ATP-sensitive K^+ (K_{ATP}) channel.**
- **Phosphodiesterases (PDEs):** each signalling episode raises cAMP or cGMP only **two- to threefold**; decline is rapid, even with continued stimulus. PDEs hydrolyse **cAMP → AMP and cGMP → GMP**; rising cyclic nucleotide **feeds back to increase PDE activity/expression**.
- **11 families (PDE1–PDE11) from 21 genes**; isoforms by alternative splicing or promoters. **Multigene families: PDE1, 3, 4, 7, 8**. Single-gene families: PDE2, 5, 9, 10, 11.

PHOSPHODIESTERASE SUBSTRATE SPECIFICITY

Substrate	Families
cAMP and cGMP	PDE1, 2, 3, 10, 11
cAMP only	PDE4, 7, 8
cGMP only	PDE5, 6, 9

- **All PDEs are in the corpus cavernosum except PDE6** (photoreceptor-specific).
- **PDE5 is the principal terminator of cavernous cGMP signalling.**
- PDE5 expression rises in **serum-starved corporal cells and high-fat-diet mice**; enhanced PDE5 activity blunts erection; **PDE5 inhibition is highly effective**.
- **PDE5 inhibition may also raise cAMP**: accumulated cGMP **competitively inhibits PDE3**, reducing cAMP breakdown. **Direct PDE3 inhibition also promotes erection**.
- **Fig. 71.8 — Immunohistochemistry (×100): PDE5 staining in cavernous smooth muscle, nerve and blood vessel wall.**

Caveolae

- **Invaginated plasma-membrane microdomains** rich in **eNOS and caveolins**, cholesterol, sphingolipids, glycosylphosphatidylinositol-linked proteins; also seven-transmembrane receptors, G proteins, adenylyl cyclase, phospholipase C, protein kinase C, Ca^{2+} pumps and channels.
- Rodents: erection appears to need **association of sGC with endothelial caveolin-1**. **Caveolin-1 falls** in aged rats, after bilateral cavernous nerve injury, and in diabetic mice.
- **Trap: in a fructose-streptozotocin diabetic rat model, caveolin-1 rose** with reduced eNOS activity — relationship complex, **species- and cause-dependent**.

Ion channels

- **Hyperpolarisation → relaxation**, mediated partly by membrane ion channels.

THE FOUR MAJOR ION-CHANNEL TYPES

Type	Opens in response to	Example
External ligand-gated	Specific extracellular molecule	Acetylcholine
Internal ligand-gated	Intracellular molecule	ATP
Voltage-gated	Membrane potential change	Na^+ , K^+ , Ca^{2+} channels
Mechanically gated	Mechanical pressure	—

- **Calcium:** smooth muscle has **neither T-tubules nor a well-developed sarcoplasmic reticulum → extracellular Ca^{2+} is the primary source**, entering via channels during action potentials; intracellular store mobilisation contributes to induced contraction.

- **Inflow regulators:** Ca^{2+} channels. **Exit regulators:** Ca^{2+} - Na^{+} exchanger and Ca^{2+} -ATPase.
- **Four K^{+} channel subtypes** in cavernous muscle: Ca^{2+} -activated (K_{Ca}), ATP-sensitive (K_{ATP}), delayed rectifier, fast transient A current. Opening $\rightarrow \text{K}^{+}$ efflux $\rightarrow$ hyperpolarisation $\rightarrow$ relaxation, even when NO and prostaglandin synthesis are blocked.
- K^{+} channels activated by **endogenous PKA, PKG or cGMP**. K_{Ca} activation enhances PDE5I relaxation in human diabetic tissue (endothelial mechanism).
- **Pharmacological openers of K_{ATP} and large-conductance K_{Ca} (Maxi-K) relax human corpus cavernosum.**
ACh relaxation of penile arteries is **blocked by large-conductance K^{+} channel inhibition**; direct activation of small-, intermediate- and large-conductance channels dilates penile arteries (partly NO-mediated, partly endothelium-independent).
- **Ca^{2+} -activated chloride channels** maintain **spontaneous tone and adrenaline-induced contraction**; activity enhanced in diabetic rabbit ED.

Additional cyclic adenosine monophosphate–signalling pathways

- **Adenosine** — released with **raised metabolic rate**; vascular actions greatest when O_2 demand is high.
Direction depends on receptor distribution:

ADENOSINE RECEPTOR SUBTYPES

Receptor	G protein	Effect
A1	Gi , Go	Inhibit adenylyl cyclase, activate phospholipase C $\rightarrow$ vasoconstriction
A3	Gi , Gq	
A2	Gs	Stimulates adenylyl cyclase $\rightarrow$ vasorelaxation

- **Role in physiological erection unclear**, but **excess adenosine with $\uparrow$ A2B signalling causes prolonged erection** in adenosine deaminase-deficient and **sickle cell transgenic mice** — **accepted priapism models**.
- Adenosine-receptor-deficient mice: **reduced erection** to nerve stimulation (attenuated fall in phosphorylated MLC) and **higher serum noradrenaline**.
- **Calcitonin gene-related peptide (CGRP) family** — CGRP, amylin, adrenomedullin; **potent vasodilators from perivascular nerves**; act via calcitonin receptor-like receptor (G protein-coupled receptor, GPCR). CGRP enhanced erection in **aged and diabetic rats**; mechanism **$\uparrow$ corporal cAMP**.
- **Prostaglandins (PGs)** — eicosanoids acting through GPCRs; at least nine receptor subtypes with splice variants.

PROSTAGLANDIN RECEPTORS — RELAXANT VERSUS CONTRACTILE

Class	Receptors	G protein	Signal
Relaxant	IP, DP1, EP2, EP4	Gas	$\uparrow$ adenylyl cyclase $\rightarrow$ $\uparrow$ cAMP
Contractile	EP1, FP, TP	Gaq	Phospholipase C $\rightarrow$ $\uparrow$ Ca^{2+} (no cAMP)
Contractile	EP3	Gai	Inhibits adenylyl cyclase $\rightarrow$ $\downarrow$ cAMP

- Corpora produce **$\text{PGF}_{2\alpha}$, PGE_2 , PGD_2 , PGI_2 and TXA_2** .
- **Contract corporal tissue:** **$\text{PGF}_{2\alpha}$, PGI_2 , TXA_2** . Relax: **PGE_1 and PGE_2** (not PGI_2).
- **Trap:** **PGI_2 (prostacyclin)** is the **predominant vasorelaxant in blood vessels** but is **contractile or neutral in erectile tissue** — explained by receptor distribution.

- Physiological role of PGs unclear; **erectogenic effect of PGE₁ as a drug extensively documented.**
- **Vasoactive intestinal peptide** — penis richly supplied with VIP and VIP-related peptides (pituitary adenylate cyclase-activating polypeptide); most such nerves **co-contain NOS**. Receptors VPAC1 and VPAC2 (GPCRs); **VPAC2 but not VPAC1 mRNA** in rat cavernous muscle. **Physiological role in human erection appears minimal**; VIP influences human vaginal smooth muscle tone.

Molecular oxygen as a modulator of penile erection

- **Partial pressure of oxygen (Po₂)** — flaccid: venous level (~35 mm Hg); erect: ~90 mm Hg.
- **O₂ is a NOS substrate** with L-arginine: **low flaccid O₂ inhibits NO synthesis**; the higher erect level induces it. **Minimum cavernous Po₂ for full NOS activity: 50–60 mm Hg.**
- **Prostaglandin H synthase (cyclooxygenase)** also uses O₂: low tension suppresses, high tension facilitates PG production.
- **Endothelin synthesis is promoted by low O₂ and inhibited by high O₂** (as in erection).

Intercellular communication

- **Gap junctions** between adjacent smooth muscle cells pass ions (Ca²⁺) and second messengers. Major component **connexin-43**, a membrane-spanning protein. Most likely explanation for **synchronised corporal relaxation and contraction**.

Concept — Every relaxant pathway, however it starts, converges on lowering cytosolic Ca²⁺ or releasing MLCP; the cyclic nucleotides differ in what makes them and what degrades them, not in what they do — and oxygen itself acts as a switch that locks the flaccid state in constrictors and the erect state in NO.

Intracavernous Tissue Architecture and Erectile Response

- **Trabeculae** provide structural support and regulation for endothelial-lined sinusoids and conduct vessels and nerves. **Relaxation → sinusoidal filling; recoil → expulsion to emissary veins → flaccidity.**
- **Decreased smooth muscle and elastin content → impaired erection.**
- **Sonic hedgehog (SHH)** — regulates organogenesis, cell division and some cancers; present in penis. **Inhibition in adult rats → rapid atrophy and disorganisation of corpus cavernosum.** SHH stimulates **vascular endothelial growth factor (VEGF)**, **NOS** and **brain-derived neurotrophic factor (BDNF)**; administration improves erection and histology in animals.

TABLE 71.2 KEY MOLECULES INVOLVED IN PHYSIOLOGIC REGULATION OF CAVERNOUS SMOOTH MUSCLE

Molecule	Function
Contraction	
High cytosolic Ca ²⁺	Binds calmodulin → activates MLC kinase
MLC kinase	Converts MLC to its active, phosphorylated form
Phosphorylated MLC	Myosin cross-bridge cycling along actin → contraction (vasoconstriction)
MLC phosphatase (MLCP)	Dephosphorylates MLC to inactive form → reverses contraction
Rho-kinase (Ca ²⁺ sensitisation)	Inhibits MLCP → maintains contraction
Relaxation	
NO	Binds sGC → cGMP
cGMP	Activates PKG
PKG	Opens K ⁺ channels, closes Ca ²⁺ channels

Molecule	Function
Low cytosolic Ca^{2+}	Ca^{2+} dissociates from calmodulin → relaxation (vasodilation)

- Table tests the **single substrate MLC_{20}** : contraction pathways phosphorylate it or block its phosphatase; relaxation pathways reverse this.
- **Trap: "MLCP" means the phosphatase**, not phosphorylated MLC.

KEY POINTS: SMOOTH MUSCLE PHYSIOLOGY

- **Relaxation of cavernous smooth muscle is the key event** in erection.
- **NO from nNOS in cavernous nerve terminals initiates** erection; NO from nNOS and **endothelial eNOS** maintains it.
- NO stimulates **cGMP** production via guanylate cyclase; cGMP activates **PKG**, which opens K^+ channels and closes Ca^{2+} channels.
- **Low cytosolic Ca^{2+} favours relaxation.**
- Contraction returns when **PDE degrades cGMP** and Ca^{2+} returns to baseline.

Concept — Corporal tone is a tug-of-war over one phosphorylation site: Ca^{2+} -calmodulin-MLC kinase and Rho-kinase keep MLC_{20} phosphorylated, and NO-cGMP-PKG removes the phosphate; ED biology is largely the story of that balance tipping toward contraction and fibrosis.

Neuroanatomy and Neurophysiology of Penile Erection

Spinal Centers and Peripheral Pathways

- Innervation is **autonomic (sympathetic and parasympathetic) and somatic (sensory and motor)** (Fig. 71.10). **Sympathetic and parasympathetic fibres merge to form the cavernous nerves**, entering corpora cavernosa and spongiosum.
- **Somatic nerves**: mainly sensation, but also **motor modulation of bulbocavernosus and ischiocavernosus** during rigid erection and ejaculation.

Autonomic pathways

SYMPATHETIC VERSUS PARASYMPATHETIC OUTFLOW

	Sympathetic	Parasympathetic
Origin	T11-L2 (in humans most often T10-T12)	S2-S4 intermediolateral cell columns
Course	White rami → sympathetic chain ganglia → lumbar splanchnic nerves → inferior mesenteric and superior hypogastric plexuses → hypogastric nerves → pelvic plexus ; penile-projecting chain ganglion cells in sacral and caudal ganglia	Preganglionic fibres in pelvic nerves → pelvic plexus , joined by sympathetic fibres from superior hypogastric plexus
Stimulation	Detumescence	Erection (pelvic plexus and cavernous nerves)

- **Cavernous nerves are branches of the pelvic plexus.** Cadaveric dissection: **medial branch accompanies the urethra**; **lateral branch pierces the urogenital diaphragm 4–7 mm lateral to the sphincter**; multiple communications between cavernous and dorsal nerves.
- **Extramural pelvic ganglion cells** lie in and along the nerves and viscera — at the **bladder-prostate junction**, **dorsal seminal vesicles** and **along the prostate** — with wide individual variation.

- Animals: **removing the cord below L4/L5 abolishes reflex erection to genital contact**, yet erection still follows exposure to an oestrous female, **medial preoptic area (MPOA) stimulation** or central **apomorphine** (dopamine agonist).
- Humans with **sacral spinal cord injury (SCI)** often keep **psychogenic** erection though reflexogenic erection is lost; commoner with **lower motor neuron lesions below T12** → centrally mediated erection works largely by **inhibiting tonic sympathetic tone**. **Psychogenic erection is usually absent with lesions above T9**. Tumescence is often marginal, insufficient for intercourse.

Somatic pathways

- Afferent sequence: **receptors in penile skin, glans, urethra and corpus cavernosum** → **dorsal nerve of the penis** → **pudendal nerve** → **S2–S4 roots** → **central grey of lumbosacral cord** → **spinothalamic and spinoreticular tracts** → **thalamus and sensory cortex** (pain, temperature, touch).
- Glans: **free nerve endings are the commonest structure** (touch, temperature), on thinly myelinated **Aδ and C fibres**; **pressure-sensitive corpuscular receptors** at much lower density.
- **Dorsal nerve: two to six branches**; in most men branches **perforate the tunica to connect with the corpus cavernosum**. Though mostly somatic sensory, it contains **nNOS-positive fibres**, reduced after cavernous nerve injury (rat); L4–L5 sympathetic chain stimulation evokes a dorsal nerve discharge → **dorsal nerve has somatic and autonomic components** regulating erection and ejaculation.
- **Onuf's nucleus (S2–S4)** — centre of somatomotor penile innervation → sacral nerves → pudendal nerve → **ischiocavernosus and bulbocavernosus**.
 - **Ischiocavernosus contraction** → **rigid-erection phase**.
 - **Rhythmic bulbocavernosus contraction** → **ejection phase of ejaculation**; antegrade flow also needs **external sphincter relaxation** and spongiosal compression.
- Animals: **brainstem sympathetic centres (A5 catecholaminergic group, locus coeruleus) directly innervate sacral motoneurons** — likely drive rhythmic perineal contraction at ejaculation. Oxytocinergic and serotonergic innervation of lumbosacral erection and perineal nuclei shown in rats.
- **Bulbocavernosus reflex** — best-known glans-evoked spinal reflex; **basis of genital neurological examination and electrophysiological latency testing**, but its **value in sexual dysfunction assessment is unclear**.

TABLE 71.3 SPINAL REFLEXES INVOLVED IN STIMULATION OF PENILE DORSAL NERVE

Stimulus	Spinal centre	Efferent	Effect
Noxious, abrupt	Sacral motor neurons	Pudendal (motor)	Bulbocavernosus reflex
Low-intensity continuous (vibratory, manual)	Sacral parasympathetic neurons and interneurons	Pelvic nerves; cavernous nerve	Detrusor inhibition and bladder-neck closure; penile erection
High-intensity continuous	Sacral motor and parasympathetic + thoracolumbar sympathetic	Pudendal, pelvic and cavernous	Ejaculation

- Pattern: **stimulus intensity and duration recruit progressively more spinal centres** — reflex twitch, then erection, then ejaculation once sympathetic outflow is added.

Concept — *Parasympathetic sacral outflow makes the erection, somatic sacral outflow makes it rigid and ejects semen, and thoracolumbar sympathetic outflow ends it — which is why lesion level predicts which type of erection survives.*

Supraspinal Pathways and Centers

- **Positron emission tomography (PET) and functional magnetic resonance imaging (fMRI)** map activation and deactivation during erotic versus neutral stimuli. Psychogenic ED, premature ejaculation and orgasmic

dysfunction may reflect altered higher function. Studies are well designed but **limited by individuality of human sexual response**.

- Integration of **visual, olfactory, imaginative and genital inputs** supraspinally is essential to initiate and maintain erection. **T8 spinal transection** study: genital afferents project to the **medullary reticular formation** via bilateral dorsal, dorsolateral and ventrolateral white matter.
- Animal supraspinal control: **limbic system** (olfactory nuclei, MPOA, nucleus accumbens, amygdala, hippocampus) and **hypothalamus** (paraventricular and ventromedial nuclei). **Key structures: medial amygdala, MPOA, paraventricular nucleus (PVN), periaqueductal grey, ventral tegmentum**.
- **Arousal**: ↑ flow in lateral hypothalamus and anterior middle cingulate cortex. **Detumescence**: ↑ flow in anteroventral hypothalamus and subgenual anterior cingulate cortex.
- **Meta-analysis core circuit of male arousal**: **cognitive** (parietal cortex, anterior cingulate, thalamus, insula), **emotional** (amygdala, insula), **motivational** (precentral gyrus, parietal cortex), **physiological** (hypothalamus/thalamus, insula); plus visual association cortices.
- **Deactivation of temporal and parietal lobes** during arousal — areas of introspection and planning → **disinhibition** before sexual activity.

TABLE 71.4 BRAIN CENTERS INVOLVED IN SEXUAL FUNCTION

Level	Region	Function
Forebrain	Medial amygdala, stria terminalis	Control sexual motivation
	Pyriform cortex	Inhibits sexual drive — hypersexuality when destroyed
	Hippocampus	Involved in erection
	Right insula, inferior frontal cortex, left anterior cingulate	↑ activity in visually evoked arousal
Hypothalamus	Medial preoptic area	Partner recognition; integrates hormonal and sensory cues
	Lateral preoptic area	Controls nocturnal penile tumescence (rats)
	Paraventricular nucleus	Facilitates erection — oxytocin neurons to lumbosacral autonomic and somatic efferents
Brainstem	Nucleus paragigantocellularis	Inhibits erection — serotonin neurons to lumbosacral neurons
	A5 catecholaminergic group, locus coeruleus	Major noradrenergic centre
Midbrain	Periaqueductal grey	Relay for sexually relevant stimuli

- Pattern: **hypothalamic PVN is the pro-erectile oxytocin output; brainstem paragigantocellularis is the serotonergic brake**.

TABLE 71.5 COMMON BRAIN ACTIVATION REGIONS WITH VISUAL SEXUAL STIMULI

Region	Functional association
Bilateral inferior temporal cortex (right > left)	Visual association
Right insula	Somatosensory information with motivational state
Right inferior frontal cortex	Sensory processing
Left anterior cingulate	Autonomic and neuroendocrine control
Right occipital gyrus	Visual processing
Right hypothalamus	Male copulatory behaviour

Region	Functional association
Left caudate (striatum)	Attention; responsiveness to new stimuli

- Regions replicated across multiple studies; **predominantly right-sided** except anterior cingulate and caudate.
- **Hypoactive sexual desire: left gyrus rectus (medial orbitofrontal cortex) stays active** during erotic stimuli, whereas it deactivates in normal desire → persistent inhibition of motivated behaviour.
- **Psychogenic ED**: extended activation of cingulate gyrus, frontal mesial and frontal basal cortex — **normalised by apomorphine**. Also ↓ **connectivity in the default mode network** (wakeful rest, attention to self and others) and **salience network** (which cues merit attention) → **aberrant appraisal of erotic stimuli and poor awareness of bodily state**.
- Cerebrovascular injury may disrupt areas integrating external stimuli with visceral arousal.

THE THREE TYPES OF ERECTION

	Psychogenic	Reflexogenic	Nocturnal
Trigger	Audiovisual stimuli or fantasy	Tactile genital stimulation	Rapid eye movement (REM) sleep
Mechanism	Brain modulates spinal centres (T11–L2, S2–S4), chiefly by suppressing sympathetic tone	Impulses reach spinal centres; some ascend (sensation), others activate autonomic nuclei → cavernous nerves	Pontine reticular formation triggers REM; cholinergic lateral pontine tegmentum active; locus coeruleus (adrenergic) and midbrain raphe (serotonergic) silent
Survives	Lower (sacral) lesions; absent above T9	Upper SCI	—

- REM PET: ↑ pontine area, amygdalae, anterior cingulate; ↓ prefrontal and parietal cortex.
- Rat: REM-sleep erections from **dorsal and intermediate lateral septum**; waking erections from **ventral lateral septum**.
- **Orgasm/ejaculation PET** — ejaculation versus stimulation without orgasm:

TABLE 71.6 BRAIN CENTERS OF ORGASM

Activity	Area	Relevance
↑ Primary	Mesodiencephalic transition zone (incl. ventral tegmental area)	Reward centre, also activated by opioids (heroin)
↑ Secondary	Midbrain lateral central tegmental field, zona incerta, subparafascicular nucleus, ventroposterior/midline/intralaminar thalamic nuclei; lateral putamen and claustrum; Brodmann 7/40, 18, 21, 23, 47 — right side only	—
↑ Other	Cerebellum (remarkably strong)	Emotional processing
↓	Amygdala and adjacent entorhinal cortex	—

- Neurobiology of orgasm, ejaculation and satisfaction still incompletely understood.

Concept — Arousal is as much the release of inhibition as the addition of drive: erotic stimuli activate reward and limbic circuits while switching off self-reflective and inhibitory cortex, and several dysfunctions map onto a failure of that switch-off.

Neurotransmitters

Peripheral neurotransmitters and endothelium-derived factors facilitating erection

- **NO from nitrergic neurons and endothelium is the principal pro-erectile transmitter** (nerve NO initiates, nerve plus endothelial NO maintains).

- Other endothelium-derived relaxants: **CO**, **H₂S**, **endothelium-derived hyperpolarising factor (EDHF)**, **PGI₂**, and **endothelin acting on endothelin-B receptors**.
- **Acetylcholine**: released on electrical field stimulation; receptors on cavernous tissue and endothelium; **not the predominant transmitter**. Acts indirectly by **presynaptic inhibition of adrenergic neurons** and **stimulation of endothelial NO release**. **Trap: atropine does not abolish neurally induced erection**.

Peripheral neurotransmitters and endothelium-derived factors opposing erection

- **Flaccidity (semicontracted muscle) = three factors: intrinsic myogenic activity, adrenergic neurotransmission, endothelium-derived constrictors** (angiotensin II, PGF_{2α}, endothelin-1).
- **Detumescence** = resumption of those factors + **cessation of NO release + PDE breakdown of cGMP + sympathetic discharge at ejaculation**.
- **Noradrenaline is the principal transmitter of flaccidity**. Contraction via Ca²⁺ entry through membrane channels; sensitisation via **protein kinase C, tyrosine kinases and Rho-kinase**.

PAIRED RECEPTOR CONTRASTS IN THE FLACCID MECHANISM

Receptor	Location/coupling	Effect
α ₁ -adrenergic	Postjunctional	Contraction
α ₂ -adrenergic	Prejunctional	Downregulates release of noradrenaline and NO
Endothelin-A	Smooth muscle	Slow-developing, long-lasting contraction; potentiates catecholamines
Endothelin-B ₁	Endothelium	Relaxation
Angiotensin II / AT1	Vasoconstrictor, proliferative arm of renin-angiotensin system (RAS)	↑ RhoA/Rho-kinase, ↑ Ca ²⁺ , oxidative stress
Angiotensin-(1-7) / Mas (GPCR)	Vasodilator, antiproliferative arm	Pro-erectile

- **Endothelin-1** — potent vasoconstrictor and mediator of detumescence; **endothelin-A blockade ameliorates penile haemodynamics in obese animal models**.
- **Constrictor prostanoids** (PGI₂, PGF_{2α}, TXA₂) synthesised by cavernous tissue; **corporal contraction mediated by TP (TXA₂) receptors**.
- **RAS**: angiotensin II and streptozotocin diabetes ↑ **toll-like receptor 4 (TLR4)**; TLR4 inhibition attenuated adrenergic contraction, preserved NOS-dependent relaxation and improved erection in vivo → **immune modulation and inflammation** may alter penile haemodynamics.
- **Transforming growth factor-β (TGF-β)** — corporal smooth muscle apoptosis, oxidative stress, ↓ **smooth muscle-to-collagen ratio**; drives impaired haemodynamics in diabetic rats via **Smad2**; ↑ RhoA/Rho-kinase, ↑ plasminogen activator inhibitor-1, ↑ profibrotic **Wnt10** pathway.
- **Tyrosine kinases** contract small penile arteries. A **tyrosine kinase inhibitor** in nerve-injured rats preserved nonadrenergic noncholinergic (NANC) fibres, downregulated sympathetic tyrosine hydroxylase fibres, improved smooth muscle-to-collagen ratio and intracorporal pressure — possibly via **large-conductance K⁺ channel activation** (seen in human tissue in vitro).

Interactions among nerves and neurotransmitters

- Marked cross-talk between pro- and antitumescence pathways, modulated by health.

- **↑ NOS activity/NO release: molecular O_2 , androgen, long-term L-arginine, repeated intracavernous PGE_1 .**
- **↓ NOS activity: castration, denervation, hypercholesterolaemia, diabetes mellitus (DM)** — partly via prejunctional α_2 inhibition of NO release.
- Isoforms regulate each other: **nNOS suppresses iNOS** after intrapenile TGF- β 1, and eNOS compensates in nNOS-knockout mice.
- **PKA and PKG cross-activate** only at **~tenfold higher** concentration of the alternate nucleotide; **PKG activation (by cGMP and cAMP) is the predominant route** by which cyclic nucleotides lower Ca^{2+} .

Molecules that modulate penile erection via the central nervous system

- Major central modulators: **dopamine, oxytocin, NO, noradrenaline, serotonin (5-hydroxytryptamine, 5-HT), prolactin**. In general, **central dopaminergic and adrenergic receptors promote, 5-HT receptors inhibit**; androgens modulate central (CNS) control.
- **Dopamine** — cell bodies in **ventral tegmentum, substantia nigra, hypothalamus**; receptors D1–D5. **D2, D3, D4, nNOS and oxytocin are co-expressed in oxytocinergic neurons of PVN and MPOA.**
 - **Apomorphine (D1 and D2 agonist) induces erection without sexual arousal.**
 - Rat PVN: dopamine agonists act on **D2 (not D3 or D4) → Ca^{2+} influx in oxytocinergic neurons → NO → oxytocinergic transmission to extrahypothalamic areas and cord → erection and yawning.**
 - **D4 stimulation → erection without yawning**; D4 plays only a modest role.
 - Dopamine and agonists **suppress prolactin** (including pituitary action).
- **Oxytocin** — neural hormone, mainly **posterior pituitary**; also neurotransmitter from PVN, hippocampus, ventral tegmental area, bed nucleus of stria terminalis, median preoptic area, spinal cord. **Serum levels rise during sexual activity.** Potent central erection inducer; **PVN most sensitive.** Release after PVN dopamine stimulation drives appetitive and reinforcing effects. **NOS inhibitors block both apomorphine- and oxytocin-induced erection → NO is the common downstream pathway.**
- **Central NO** — NO and sGC throughout the brain incl. PVN; activity **declines with ageing and experimental diabetes; testosterone enhances apomorphine erections and MPOA NOS** (dopamine-responsive NOS; NO-enhanced dopamine release).
- **Serotonin** — cell bodies in **midline raphe nuclei**; projections to hypothalamus, limbic system, neocortex, cord. Generally **inhibits desire and erection**, but subtype-dependent: **5-HT1A and 5-HT2 inhibit erection yet facilitate ejaculation; 5-HT2C facilitates erection (rats)**; lumbosacral 5-HT2C receptors may be downstream of dopamine–oxytocin and melanocortin pathways.
- **Noradrenaline** — cell bodies in **locus coeruleus and A5 group** (pons, medulla); ascend to PVN, supraoptic and periventricular nuclei, thalamus, neocortex; descend to cord and cerebellum. **Centrally stimulatory — opposite of its peripheral action.** Clonidine (α_2 agonist) **↓ sexual behaviour**; yohimbine (α_2 antagonist) **weakly improves erection.**
- **Melanocortins** — **melanocortin-4 receptor (MC4R)** in CNS, pelvic ganglia and glans nerve fibres; controls food intake and energy expenditure; **MC4R agonist enhanced copulation and erection in mice.**
- **Prolactin** — **hyperprolactinaemia suppresses sexual function** via **↓ MPOA dopaminergic activity and ↓ serum testosterone.**
- **γ -Aminobutyric acid (GABA)** — PVN GABA inhibits pro-erectile signalling. **Baclofen (GABA_B agonist)**, systemic or lumbosacral intrathecal, reduces erection in rats; **GABA_A activation blunts response** to dopamine agonists and oxytocin.

- **Opioids** — dose- and site-dependent: **small MPOA morphine doses induce sexual behaviour; larger doses inhibit** oxytocin- and dopamine-driven erection and yawning; PVN morphine prevents noncontact erections and impairs copulation (possibly NO-mediated).
- **Cannabinoids** — **CB1 activation inhibits** via PVN oxytocinergic neurons; **CB1 antagonism in PVN induces erection** via glutamate and NO.

TABLE 71.7 CENTRAL NEUROTRANSMITTERS AND THEIR FUNCTION

Transmitter	Receptor and function
Dopamine	D1 and D4 — enhance erection; D2 — enhances seminal emission (and, in the PVN, drives oxytocin-mediated erection)
Serotonin	5-HT overall — inhibits sex drive and spinal sexual reflex; 5-HT1A — inhibits erection, facilitates ejaculation; 5-HT2C — enhances erection
Noradrenaline	Enhances sexual function (central)
GABA	Inhibits erectile signals
Opioids	Inhibit erection
Cannabinoids	Inhibit sexual function
Oxytocin	Enhances appetitive and reinforcing effects
NO	Mediates erection at PVN
Melanocortins	MC4R — enhances erection
Prolactin	Suppresses sexual function

- Pattern: **promoters — dopamine, oxytocin, NO, central noradrenaline, melanocortins, 5-HT2C; brakes — 5-HT (most subtypes), GABA, opioids (high dose), cannabinoids, prolactin.**
- Trap: noradrenaline is pro-sexual centrally but the principal antierection transmitter peripherally.

KEY POINTS: PHYSIOLOGY OF PENILE ERECTION

- **Neuronal integrity at brain, cord and peripheral levels** is essential for modulating erection.
- Many neurotransmitters modulate arousal and erection — key examples **dopamine, 5-HT, oxytocin, NO, noradrenaline.**
- Central integration of erotic stimuli is complex; **MRI and PET** have given new insight.

Concept — Central control converges on PVN oxytocinergic neurons with NO as the shared downstream step: dopamine and melanocortins push that output, while serotonin, GABA, opioids, cannabinoids and prolactin brake it — the template for both centrally acting drugs and drug-induced dysfunction.

PATHOPHYSIOLOGY OF ERECTILE DYSFUNCTION

- Section epigraph: a humorous verse on ageing and the loss of spontaneous erection — framing **age as the dominant theme.**
- **Definition: ED is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction** (American Urological Association, AUA, 2018).
- "ED" replaces the antiquated and pejorative "impotence".

Incidence and Epidemiology

- **Prevalence rises with age. Kinsey report (1948):** ED in 2% at age 40 and 25% by age 65; confirmed by later, more rigorous studies.

- **ED is not only a condition of older men** — mild or moderate/severe ED reported in a meaningful minority of US men aged 18–31.
- **Subjective bother declines with age** even as prevalence rises.
- Comparable prevalence in other nations and regions.

THE MAJOR POPULATION STUDIES OF ED

Study	Design and population	ED measure	Bottom line
National Health and Social Life Survey (NHSLs)	US national probability survey, 1992; men and women aged 18–59; broad sexual-practice enquiry	Single item	Prevalence rises with each decade from the 20s to the 50s
Massachusetts Male Aging Study (MMAS) — landmark	First cross-sectional, community-based, random-sample, prospective study; 1709 men aged 40–70 , Boston; 1987–89, resurveyed 1995–97	Nine items; extensive physiological and psychosocial data	Incidence per 1000 man-years: 12.4 (40s), 29.8 (50s), 46.4 (60s). Age-adjusted incidence higher with DM (50.7), treated heart disease (58.3), treated hypertension (42.5)
Multiethnic Study of Atherosclerosis (MESA)	Deliberately diverse US sample, addressing the homogeneity of NHSLs and MMAS	Single item	ED present in all groups; ED independently predicted incident major coronary heart disease events (hazard ratio ~1.9)

- Series tests two ideas: **ED is age-related but comorbidity-driven**, and **ED predicts cardiovascular events**.

Concept — *Age is the strongest demographic predictor, but the conditions that accumulate with age — diabetes, heart disease, hypertension — carry more risk than age itself, and ED is an early readout of the same vascular disease.*

Risk Factors

- Associated conditions: **poor health/chronic disease, vascular and metabolic disease, nervous system disorders, hormonal disorders, medications, psychiatric/psychological disorders, substance abuse, sociodemographic factors** (Fig. 71.12).
- Sociodemographic link may reflect comorbidity (more common with lower socioeconomic status), differing expectations of sexual response, or unknown factors.
- **Unifying theme: vascular and endothelial dysfunction** — directly (vascular, metabolic disease) or indirectly (via neural innervation or hormonal milieu).

TWO WAYS OF CLASSIFYING ED

By cause	By neurovascular mechanism
Diabetic, iatrogenic, traumatic, etc.	Failure to initiate (neurogenic); failure to fill (arterial); failure to store (venous)

- **Fig. 71.12** — Functional classification: **arterial** (arterial, arteriolar), **cavernosal** (tunica, cavernous muscle, gap junction, endothelium, fibroelastic trabeculae, emissary vein), **neurologic** (sensory, motor, autonomic, neurotransmitters), **hormonal** (testicular, pituitary, thyroid), **psychologic**, systemic diseases and drugs. **ED rarely derives from one source; most cases have a psychogenic component** (Carrier, Brock, Kour et al., 1993).
- **Fig. 71.13** — **NOS uncoupling**: oxidative stress → tetrahydrobiopterin (BH4) oxidation, increased S-glutathionylation and NOS dimer disruption → uncoupled NOS; arginase activity → arginine depletion → uncoupling; uncoupled NOS and NADPH (reduced nicotinamide adenine dinucleotide phosphate) oxidase generate superoxide → combines with NO → **peroxynitrite** → **vascular dysregulation**. Potential inhibitors: **antioxidants and NADPH oxidase inhibitors; folic acid (protects BH4); arginase inhibitors**.

Psychosocial

- **A psychogenic component is common and possibly ubiquitous**, as primary cause or secondary contributor. Hypothalamus, limbic system and cortex send stimulatory or inhibitory messages to spinal centres.
- **Two proposed mechanisms of inhibition: (1) direct CNS/spinal inhibition of erection; (2) excessive sympathetic activity or raised peripheral catecholamines** ↑ smooth muscle tone and prevent relaxation.
- CNS evidence: depressed animals have ↓ **serum dopamine and penile dopamine receptors**; **nucleus accumbens dopamine agonist** improves intromission and mount latency in stressed rodents. fMRI in psychogenic ED: ↓ **activity in right anterior insula and right orbitofrontal cortex** (abnormal cognitive and motivational processing); aberrant network connectivity linked to **state anxiety**. **Clinical salience still ambiguous**.
- Peripheral evidence: sympathetic stimulation or systemic adrenaline **detumesces** the erect rodent penis. Chronic water-immersion stress: ↑ response to noradrenaline, ↑ **cavernous Rho-kinase-1**, ↓ **smooth muscle-to-collagen ratio**. **Stress from adulthood worsened the ratio, stress since pre-puberty did not**.
- **Men with psychogenic ED have higher serum noradrenaline** than controls or men with vasculogenic ED.
- **Social factors**: absent partner, partner conflict, differing expectations/mores by demographics (including racial identification), culture, beliefs and sexual orientation.
- **Pornography: self-reported problematic use is associated with ED, but use per se is not clearly linked** — impact is individual.
- **Post-traumatic stress disorder** ↑ ED risk. These social and mental-health factors act through psychogenic mechanisms.

Neurogenic

- Erection is neurovascular: **any disease, dysfunction or injury (including surgical) of brain, cord or peripheral erectogenic nerves (pudendal, cavernous)** can cause ED. Neuropathy **does not exclude other causes**; confirming a neurogenic cause is difficult.
- Implicated brain regions: caudate, dorsal anterior cingulate, anterior insula, prefrontal areas, median preoptic area, PVN, thalamus, hippocampus.
- **Central disorders**: Parkinson disease, stroke, encephalitis, temporal lobe epilepsy, tumours, dementias (including Alzheimer disease), multiple system atrophy. **Mechanism: insufficiency of dopaminergic and/or oxytocinergic pathways**; oxytocin into the bed nucleus of the stria terminalis induces erection in rats (glutamatergic, nitroergic, dopaminergic).
- **Traumatic brain injury** linked to ED; in **American football players, concussion history showed a dose-response relationship with ED medication prescription**.
- **Spinal lesions**: SCI (commonest), spina bifida, disc herniation, syringomyelia, tumours, multiple sclerosis. Nature, level and extent dictate effect.

COMPLETE SPINAL CORD LESIONS — UPPER VERSUS LOWER

	Complete upper cord lesion	Complete lower cord lesion
Reflexogenic erection	Preserved in ~95% (sacral parasympathetic arc intact)	Preserved in only ~25%
Psychogenic erection	Lost	May occur — cortical inhibition of tonic thoracic sympathetic tone

	Complete upper cord lesion	Complete lower cord lesion
Quality	Brief; need continuous stimulation	Often marginal tumescence

- Cord lesions may also impair **ejaculation**; **emission** is lost after sympathetic chain injury in **retroperitoneal surgery**.
- **Iatrogenic ED after pelvic surgery** (radical prostatectomy, cystoprostatectomy, abdominoperineal resection) and **pelvic radiation is common**. Nerve-sparing modifications for rectal, bladder and prostate cancer markedly improved outcomes.
- Yet nerve injury, local circulatory disruption and architectural change mean **most men have at least transient decline; relatively few recover full baseline; baseline ED and age over 60 → greater risk**.
- **Cavernous nerve injury in animals**: ↑ neurotoxic M1 macrophages, inflammatory cytokines, RhoA/Rho-kinase/LIM kinase 2, tumour necrosis factor- α (TNF- α), angiotensin II, tyrosine hydroxylase neurons, specific long noncoding and circular RNAs; ↑ oxidative stress, fibrosis, apoptosis; ↓ **smooth muscle-to-collagen ratio**; ↓ **SHH**.
- **Pelvic fracture** → cavernous nerve and/or vascular injury. With **posterior urethral injury, early realignment has a better potency-preservation rate than delayed anastomosis**.
- **Diabetes** causes ED via neurogenic or vascular damage, both impairing NO release.
- **Age-related loss of penile tactile sensitivity** reduces sacral reflexogenic input — more important as psychogenic drive falls with age.

Endocrinologic

Androgens

- **Low serum testosterone (T) is frequent in ED**. AUA 2018: measure T in all men with ED — morning, fasting (glucose and protein intake acutely suppress androgens).
- Androgens act via the **androgen receptor (AR)**; effects on **libido and sexual behaviour well established**. In low T, supplementation ↑ interest, ↑ erection and improves non-sexual quality of life.
- **Thresholds**: T for normal nocturnal erections ~200 ng/dL; many men have symptoms (incl. loss of nocturnal erections) below ~320 ng/dL; **androgen-deficiency symptoms become progressively more frequent as T falls**.
- **AUA T guideline: biochemical low T = <300 ng/dL on a morning draw**.
- **Free T** — may predict ED where total T is borderline; **controversial and not in AUA guidance**.
- **AR CAG repeats** (exon 1): longer repeats ↓ AR transcription; weak association with ED **only in men with at least low-normal T (≥330 ng/dL)**; not replicated in all studies.
- **Low libido is the commonest sexual symptom of low T**. Older men more often symptomatic (cause or effect unclear). **Many men with low T are asymptomatic**.
- **Endocrine Society and AUA: consider T only with at least two clearly low morning levels plus symptoms clearly attributable to low T**.
- **Correlates of low T in ED**: age, DM, high total cholesterol, anaemia, waist circumference, **obesity (body mass index, BMI, ≥30 kg/m²)**.
- Low T associates with ↑ fasting insulin, glucose and glycated haemoglobin in men who develop diabetes → **hypogonadism may be a sentinel event for diabetes**; androgen deficiency links to insulin resistance, type 2 DM, metabolic syndrome (MetS) and visceral fat. **Visceral fat acts as an endocrine organ** (inflammatory cytokines, endothelial dysfunction).
- Mechanisms: T regulates **NO/cGMP**, RhoA/Rho-kinase, adrenergic tone and smooth muscle. **Castration** → ↓ arterial inflow, venous leak, ↓ nerve-evoked erection; ↓ smooth muscle, nNOS and eNOS; ↑

oxidative stress; ↓ autophagy (protective); ↑ endothelial and smooth muscle apoptosis; impaired endothelial progenitor cells; ↓ myogenic differentiation of mesenchymal progenitors. **T supplementation mostly reverses these in animals.**

- Clinically: **long-term androgen ablation for prostate cancer → poor libido and ED.** T monotherapy gives a **modest** erectile gain in hypogonadism, possibly needing sustained use; some data suggest **benefit is mainly in desire rather than erection.**

HYPOGONADISM BY LEVEL

Hypogonadotrophic	Hypergonadotrophic
Congenital, or hypothalamic/pituitary tumour or injury	Testicular tumour, injury, surgery or severe orchitis

- **Most low T occurs without marked gonadotrophin (luteinising and follicle-stimulating hormone) abnormality.**

Nonandrogens

- **Oestradiol (E2):** high E2 and/or **low T:E2 ratio** independently ↑ ED risk; excess from systemic disease and **obesity (aromatisation).** But some E2 is essential for male desire.
- **Hyperprolactinaemia** (pituitary adenoma; antipsychotics) → ↓ libido, ↓ rigidity and nocturnal erections; low T via **inhibition of hypothalamic gonadotrophin-releasing hormone.**
- **Trap: low prolactin is also harmful** — associated with MetS, arteriogenic ED, premature ejaculation and anxiety; men on dopamine agonists with **prolactin <3 ng/mL** had more ED than those normalised to **3–20 ng/mL.**

THYROID DYSFUNCTION AND ED

Hyperthyroidism	Hypothyroidism
↓ Libido (↑ circulating oestrogen); ED via ↑ sympathetic tone , disrupted NO signalling, depression	Suppressed hypothalamic–pituitary–gonadal axis, ↑ prolactin , depression, possible MetS

Concept — *Testosterone governs desire more than mechanics: it maintains the NO machinery and smooth muscle of the corpora, but its clinical deficiency shows first as low libido, which is why treatment requires both repeated low levels and attributable symptoms.*

Vascular

- **Arteriogenic** — atherosclerotic or traumatic occlusion of the **hypogastric–cavernous–helicine tree** → ↓ perfusion pressure → **longer time to maximal erection and ↓ rigidity.** Usually part of **generalised atherosclerosis.**
 - Risk factors: hypertension (HTN), hyperlipidaemia, **tobacco**, DM, blunt perineal/pelvic trauma, MetS, sedentary lifestyle, obesity, **pelvic irradiation.**
 - **Penile Doppler abnormalities associate with tobacco use, vascular disease and major cardiac events.**
 - **Focal stenosis of common penile or cavernous artery → young men after blunt pelvic or perineal trauma.**
 - Modifiable: **tobacco cessation, weight loss, regular exertion** improve erection (arteriogenic fitness).
- **Cardiovascular diseases** — ED highly prevalent with HTN, hypercholesterolaemia, coronary artery disease (CAD), peripheral vascular disease. **Five mechanistic links:** (1) endothelial dysfunction (eNOS, oxidative stress); (2) smooth muscle defects (hyperplasia, altered contractility); (3) autonomic dysregulation (neuropathy, ↓ nNOS); (4) hormonal disruption (hypogonadism); (5) metabolic defects (hyperlipidaemia, advanced glycation end products).
 - **Artery-size hypothesis: penile arteries are smaller than coronary arteries, so endothelial dysfunction shows in the penis first.** In men with CAD, ED preceded CAD in most cases by a mean of ~2 years.

- **Reactive hyperaemia (microvascular) associated more strongly with ED than flow-mediated dilation (conduit arteries)** → microvascular disease may be primary.
- **ED = early indicator of systemic endothelial dysfunction and an indication for cardiac risk stratification.**
- **Hyperlipidaemia** — ↑ low-density lipoprotein (LDL), ↓ high-density lipoprotein (HDL) predict prevalent and incident ED. **Statins associated with better erections and response to erectogenic therapy.**
Mechanisms: altered penile NOS and endothelium; **oxidised LDL inhibits NO relaxation and ↑ contractility**; oxidative stress; **endogenous NOS inhibitors (N^G-monomethyl-L-arginine, asymmetric dimethylarginine, ADMA); ↑ arginase depleting arginine**; procontractile TXA and PGs; ↓ **VEGF** impairing endothelium-dependent relaxation.
- **Obesity** — high BMI and processed/high-fat diets independently ↑ ED. Animals: ↑ endothelin receptors, oxidative stress, ↓ T, ↓ smooth muscle-to-collagen ratio, ↓ NO and ACh reactivity. **Visceral obesity** → oxidative stress and insulin resistance in penile tissue, partly via **mechanistic target of rapamycin (mTOR)**. Genome-wide study: locus on **chromosome 6 near SIM1** (melanocortin/leptin signalling), **significant independent of BMI. Caloric restriction, dietary change, lifestyle weight loss and bariatric surgery improve erectile function.**
- **Hypertension** — **independent risk factor**; higher prevalence, earlier onset, especially with **wide blood pressure variability**. In type 1 DM, risk rose with blood pressure but was mitigated by antihypertensives. Models: ↓ smooth muscle-to-collagen, ↓ NOS, ↑ α_1 expression; Rho-kinase regulates vascular Ca^{2+} sensitivity; TLR activation by damage-associated molecular patterns; endoplasmic reticulum stress in spontaneously hypertensive rats (SHR); RAS → NADPH oxidase → oxidative stress. **Rho-kinase inhibition ameliorates hypertensive ED in animals.**

Mechanism of Vascular Erectile Dysfunction

Arteriogenic erectile dysfunction

- **Cavernous O₂ tension is lower in arteriogenic than psychogenic ED.** Sequence: **penile hypoxaemia** → ↓ smooth muscle, ↑ TGF- β , ↑ collagen types I and III, partly via **hypoxia-inducible factor-1 α (HIF-1 α)**.
- HTN: ↑ arterial wall/lumen ratio and ↑ penile vascular resistance in SHR; mitochondrial damage, nerve degeneration, oxidative stress, ↑ adrenergic contractility, disrupted NOS. **Structural change is RAS-mediated** (see below).

LOWERING BLOOD PRESSURE VERSUS PROTECTING THE PENIS

Agent	Systemic BP	Penile end-organ damage
AT1 receptor blocker	Lowered	Partly prevented
Hydralazine; Ca ²⁺ channel blocker	Lowered	Not mitigated
Nebivolol (NO-potentiating β -blocker)	Lowered	Mitigated
Amlodipine	Lowered	Not mitigated

- **Trap: penile protection depends on RAS blockade or NO potentiation, not on the blood pressure reduction itself.**

Enhanced smooth muscle contraction and vasoconstriction

- **↑ RhoA/Rho-kinase → corporal hypercontractility** in DM, hypercholesterolaemia, HTN, hypogonadism, cavernous nerve injury and ageing. **Rho-kinase inhibition** ↓ pelvic atherosclerosis, improves haemodynamics, ↓ apoptosis, preserves NOS; relaxes human corporal cells.

- **Endothelin-1** ↑ in atherosclerosis, HTN, hypercholesterolaemia; ↑ in venous and cavernous blood in organic ED. **Trap: a pilot endothelin-A antagonist trial in ED was negative.**
- **Angiotensin II** restricts vasodilation via AT1; **AT1 blockers and angiotensin-converting enzyme (ACE) inhibitors show promise in ED with HTN and with atherosclerosis respectively.**
- **Maxi-K⁺ channels:** depolarisation → K⁺ efflux → L-type Ca²⁺ channel closure → ↓ Ca²⁺. Silencing raises Ca²⁺ only transiently (compensation).
- Altered **gap junction** communication; ↑ **collagen** in arteriogenic and obesity-related ED.
- **High adrenergic tone** (medical or psychological) → insufficient trabecular relaxation → **two hits: ↓ arterial inflow and impaired emissary vein coaptation.**

Impaired endothelium-dependent smooth muscle relaxation

- **Endothelial dysfunction = common link between cardiovascular disease and ED. Brachial flow-mediated dilation is impaired in ED, in proportion to ED severity.**
- **Fewer circulating endothelial progenitor cells** (bone-marrow-derived, repair endothelium) in ED with heart disease, obesity and low T.
- Hypertensive models: impaired relaxation from **angiotensin II, thromboxane, superoxide and raised intravascular pressure.**

TABLE 71.8 VASCULAR AND STRUCTURAL CHANGES LEADING TO ERECTILE DYSFUNCTION

Structure	Change in ED
Cavernous artery	↑ Vascular resistance, narrow lumen
Smooth muscle	↑ Tone (hypertonicity) ; ↓ content ; altered K ⁺ channels and gap junctions
Cavernous tissue	Fibrosis (collagen deposition, elastin loss); impaired veno-occlusion
Endothelium	Impaired endothelium-dependent relaxation
Tunica albuginea	Altered elastic and collagen fibres
Neurotransmitters	↓ nNOS, ↓ eNOS

- Table maps **every anatomical level of the erectile apparatus to its failure mode**; the common endpoint is failure to relax and failure to trap blood.

Concept — *Vascular ED is not simply narrowed pipes: hypoxia, RAS activation, Rho-kinase and endothelial failure convert the corpora from relaxant to contractile and from muscle to collagen, so inflow failure becomes storage failure.*

Corporal Structural Defects

- **Failure to restrict venous outflow (veno-occlusive dysfunction) is one of the commonest causes of vasculogenic ED.** Causes: **degenerative tunical change, fibroelastic structural alteration, insufficient trabecular relaxation, venous shunts.**
- **Tunica:** degeneration (senescence, DM) or trauma (**penile fracture**) impairs subtunical and emissary vein compression. **Peyronie surgery** may disrupt the subtunical areolar layer or tunica.
- **Priapism shunt surgery** → persistent **glans-cavernosum or cavernosum-spongiosum shunting** and venous leak; ED may also reflect **priapism-related fibrosis.**
- **Fibroelastic trabecular and smooth muscle alterations** → venous leak; seen in vasculogenic and neurogenic ED and animal nerve-injury and vascular models.
- **Sonic hedgehog protein** — regulates endothelial growth, angiogenesis, migration, neuronal differentiation and axon guidance. ↓ **SHH after cavernous nerve injury and in DM** → nerve degeneration,

corporal apoptosis, ↓ smooth muscle. **SHH replacement in nerve-injured rats restores erection**, possibly via **BDNF**.

- **Endothelium** — source of NO, EDHF, PGI₂, hydrogen peroxide; mediates flow-mediated dilation; influences mitogenesis, platelet aggregation and neutrophil adhesion — via **chemical mediators or myoendothelial gap junctions**. **Dysfunction is an early indicator of vascular disease**. Mechanisms in ED: ↓ endothelial cell density; ↓ junction proteins (**claudin-5, vascular endothelial-cadherin, platelet endothelial cell adhesion molecule-1**).

Concept — *Veno-occlusion is an emergent property of healthy tissue — an elastic tunica compressing veins against fully relaxed trabeculae — so anything that stiffens the tunica, fibroses the trabeculae or prevents relaxation presents as "venous leak".*

Medication Induced

- Medication is commonly associated with self-reported ED. Many conditions in the ED population (depression, DM, cardiovascular disease) **themselves cause ED**, and their treatment may add to it.
- Some drugs affect **desire, ejaculation or orgasm**, which patients may report as "ED" or which may secondarily impair erection. **Interpret self-reported and questionnaire drug-ED data with caution.**

Antihypertensive agents

- Almost all list ED as a side effect, but risk differs by class and **some are beneficial**. **The underlying hypertensive disease may be the principal driver** of ED in treated men.
- **Thiazide diuretics** — principal action: **inhibition of the Na-Cl cotransporter in the distal convoluted tubule** (weak carbonic anhydrase inhibition causes cellular alkalinisation and vasodilation). **Higher ED likelihood even versus other antihypertensives**; mechanism unknown.
- **β-Adrenergic blockers** — only ~10% of penile adrenoceptors are β; their stimulation mediates relaxation. In vitro this is attenuated by **nonselective propranolol** (possibly a prejunctional α₂ effect) but **not by cardioselective practolol**.
 - **Nonselective (propranolol) → more ED than ACE inhibitors.**
 - **Carvedilol** (nonselective β plus α₁-blocking vasodilation) → worsened sexual function.
 - **Nebivolol** (selective β₁ with **NO-mediated vasodilation**) → no reduction, sometimes improvement, versus metoprolol and atenolol.
- **α-Adrenoceptor blockers** — α-adrenoceptors (especially the α_{1L} subtype) are prevalent in penile vascular and cavernous tissue and promote contraction. α₁ antagonists increase/prolong cavernous relaxation; prejunctional α₂ modulates noradrenaline release. **α-Blockers for HTN/benign prostatic hyperplasia (BPH), e.g. doxazosin, are not associated with ↑ ED.**
- **α₂ agonists (clonidine) ↓ erectile function** by peripheral and central mechanisms.
- **ACE inhibitors (ACEIs)** — no evident interfering action; given the RAS role, **neutral to beneficial in most human studies**; central ACEI enhanced centrally induced erections in diabetic rats.
- **AT1 receptor antagonists (angiotensin receptor blockers, ARBs; losartan, valsartan, candesartan)** — **reverse penile vascular structural change** in hypertensive and ageing animals; **seem to improve erectile function** in humans. **Valsartan versus carvedilol crossover: valsartan improved pre-existing dysfunction with no adverse sexual effect over 12 months**; losartan for 3 months improved function.
- **Ca²⁺ channel blockers** — **no generalisable adverse effect** on erection.
- **Aldosterone receptor antagonists** — mineralocorticoid receptors in corporal tissue; **hyperaldosteronism ~threefold ED risk**. **Spironolactone**: nonselective, with progesterone and AR affinity → **↓ libido**,

gynaecomastia, ED; yet it reduces mineralocorticoid-driven corporal inflammation (TNF- α , nuclear factor- κ B, interleukin-6) → **theoretical peripheral benefit**.

TABLE 71.9 EFFECT OF ANTIHYPERTENSIVE AGENTS ON SEXUAL FUNCTION

Agent	Effect	Mechanism
Thiazide diuretic	ED	Unknown
Nonselective β -blocker	ED	Prejunctional α_2 effect
α_1 -blocker	↓ ED rate; may alter ejaculation	Loss of sympathetic closure of internal sphincter/proximal urethra and loss of seminal emission
Central α_2 -agonist (clonidine)	ED	Central α_2 action
ACEI	Possible ↓ ED	—
ARB	Possible ↓ ED	—

- **Worst: thiazides and nonselective β -blockers** (reduced by combination therapy and weight loss); preferred first agents in men with ED: α_1 -blockers, ACEIs, ARBs.

- **Trap: α_1 -blockers protect erection but disturb ejaculation.**

Psychotropic medication

- Receptor complexity and interlinked CNS pathways make effects on sexual function — positive and negative — very likely.
- **Antipsychotics** — sexual dysfunction common in comparative study; **newer agents (clozapine, risperidone) cause fewer desire and erectile deficits**. Therapeutic action: **dopamine blockade in limbic and prefrontal areas**. Sexual adverse effects: **α -adrenergic blockade, anticholinergic action, basal-ganglia antidopaminergic action**. Dopamine blockade → hyperprolactinaemia → ↓ T and ↓ permissive cerebral dopamine.
 - Rats: **D1 activation in MPOA facilitates erection** (oxytocinergic, spinal cholinergic); **D2 there may oppose**. Haloperidol and flupentixol reduce apomorphine erections via D1 antagonism.

TYPICAL VERSUS ATYPICAL ANTIPSYCHOTICS

Typical (older)	Atypical (newer)
Frequent extrapyramidal effects	Less common extrapyramidal effects
Hyperprolactinaemia more common	Less
More sexual dysfunction	Fewer desire and erectile deficits

- Differences reflect **receptor affinity and regional avidity**.
- **Antidepressants** — sexual effects drive **non-compliance**; clearly ↑ odds of ED, especially in young men.
 - **Tricyclic antidepressants (TCAs)**: inhibit catecholamine reuptake; sexual effects via **anticholinergic and α -adrenergic actions**, possibly 5-HT antagonism. **Orgasmic disorders frequent → TCAs used to inhibit ejaculation**. TCAs and monoamine oxidase inhibitors now rarely used.
 - **Selective serotonin reuptake inhibitors (SSRIs)** — commonest current antidepressants; change sexual function in a substantial proportion. Mechanisms: **5-HT₂ and 5-HT₃ stimulation inhibiting spinal erectogenic pathways**; ↓ MPOA dopamine; altered NO metabolism. Effects may persist after cessation.
 - **Mitigation: co-treatment with PDE5Is, bupropion (dopaminergic) or buspirone (5-HT_{1A})**.
 - **Paroxetine** — **high ED incidence**; citalopram — **lesser impact**. Paroxetine, not citalopram, inhibited NO production and caused ED in rats → **mechanisms beyond serotonin reuptake**.

- **Post-SSRI sexual dysfunction (PSSD):** ED, **genital anaesthesia**, altered orgasm/ejaculation persisting long after stopping; possible hormonal change, 5-HT receptor downregulation, altered 5-HT-dopamine interaction, **epigenetic change**.
- **Other antidepressants** — mirtazapine and nefazodone tend to benefit sexual function, attributed to action at the 5-HT_{2C} receptor (formerly termed 5-HT_{1C}), whose central stimulation induces erection in animals. Trazodone → isolated priapism reports, attributed to its metabolite m-chlorophenylpiperazine; trazodone ↑ nocturnal erections despite ↓ REM.
- **Anxiolytics** — benzodiazepines potentiate GABA (reticular, limbic) and may affect 5-HT and dopamine; GABAergic drugs block apomorphine erections. Lithium plus benzodiazepine → more sexual dysfunction than lithium alone. Bupropion (dopamine reuptake inhibitor) and buspirone (5-HT_{1A}) have fewer sexual effects.

ANTICONVULSANTS AND SEXUAL FUNCTION

Drug	Sexual effect
Lamotrigine	Sexual function, bioavailable T and gonadal efficiency comparable to controls; reports of improved function and hypersexuality
Carbamazepine	Lower than lamotrigine; orgasmic dysfunction common
Phenytoin	Lower than lamotrigine
Valproate	Loss of desire common

- Epileptic discharges themselves disturb hypothalamic-pituitary function and sex hormones.

Antiandrogens

- Androgens modify behaviour via CNS ARs; antiandrogens block production or receptor. Effects of deficiency vary from complete loss to normal function.
- **Nocturnal (REM) erections are androgen-dependent; erections to visual sexual stimuli are androgen-independent.**
- Peripheral: castration ↓ rat corporal NOS; T restores it, but **finasteride blocks the restoration** → dihydrotestosterone (DHT) is the key penile androgen.

ANTIANDROGEN CLASSES AND SEXUAL FUNCTION

Class	Effect on serum T	Sexual effect
5α-reductase inhibitors (5ARIs) — finasteride, dutasteride	Least effect on circulating T	Persistent, hard-to-treat ED and ↓ drive reported (especially alopecia users), persisting after cessation; longer exposure → greater risk
Nonsteroidal AR antagonists — flutamide, bicalutamide	Alone: T rises	Bicalutamide alone: smaller loss of desire than castration , but erectile loss common even at 50 mg
Steroidal — cyproterone	—	Gradual loss of desire over years (as with flutamide)
Luteinising hormone-releasing hormone (LHRH) agonist	Initial surge, then castrate	Profound loss of desire, usually with ED
LHRH antagonist	Immediate suppression	
Nonsteroidal antiandrogen + LHRH agent	Castrate range	Loss of desire is the main side effect

- **Rodent 5ARI data:** ↓ nerve-evoked erection, ↓ smooth muscle-to-collagen, ↓ eNOS, ↑ apoptosis, ↓ autophagy; **after 8 weeks (unlike 4 weeks) changes failed to resolve**, with ↑ TGF- β and Smad2/3.
- **Trap:** the antiandrogen with least effect on serum T (5ARI) is the one linked to persistent post-treatment dysfunction.

Miscellaneous drugs

- Mostly **anecdotal case reports or post-marketing alerts**, not controlled trials.
- **Statins** — some studies suggest worsening ED (possibly reflecting progression of the vascular disease that required treatment); more show **benefit**: improved nocturnal activity and questionnaire scores in hyperlipidaemia; benefit with ACEI in poor PDE5I responders; rescue of sildenafil non-response; better recovery after bilateral nerve-sparing prostatectomy with sildenafil; improved International Index of Erectile Function (IIEF). **Meta-analysis: statins improve mean IIEF.**
- **Histamine H₂ antagonists** (cimetidine, ranitidine) — single in vitro study: H₂ stimulation relaxes cavernous tissue (endothelial NO) → blockade **theoretically** contributes to ED.
- **Opiates** — **long-term intrathecal opiates → hypogonadotrophic hypogonadism** and sexual dysfunction, **reversible with supplementation**. Opioid antagonists did not improve nocturnal tumescence in older men with ED; naloxone had no effect in healthy volunteers.
- **Antiretroviral agents** — hypogonadism and ED commoner in men with human immunodeficiency virus (HIV); multisystem disease and polypharmacy make drug attribution difficult (metabolic, psychosocial factors).
- **Tobacco** — clearly linked. **Boston Area Community Health (BACH) survey: dose-response, significant at ≥20 pack-years; passive smoking** gave a small, non-significant increase comparable to ~10–19 pack-years. Mechanisms: nicotinic contraction of corporal cells (reduced by cyclooxygenase, Rho-kinase and TXA₂-synthesis inhibition); ↓ NOS, ↑ oxidative stress and apoptosis; ↓ nNOS; endogenous NOS inhibitors; ↑ **arginase I**.
- **Alcohol** — small amounts: ↑ **erection and drive** (vasodilation, anxiolysis). Large amounts: **central sedation, ↓ libido, transient ED**; dose-dependent association. Mechanisms: polyneuropathy, hormonal and metabolic effects, oxidative stress, ↑ endothelin-1 reactivity, RhoA/Rho-kinase. **Duration matters: 14-day but not 7-day ethanol exposure damaged cavernous endothelium in mice.**

TABLE 71.10 DRUG-INDUCED ERECTILE DYSFUNCTION AND SUGGESTED ALTERNATIVES

Class	Known to cause ED	Suggested alternatives
Antihypertensives	Thiazide diuretics; nonselective β-blockers	ACEIs; ARBs; selective β-blockers (vasodilating, e.g. nebivolol); α-blockers; Ca²⁺ channel blockers
Psychotropics	Antipsychotics; antidepressants; anxiolytics	Newer agents — bupropion, buspirone
Antiandrogens	AR antagonists; LHRH agonists; 5ARIs	Not applicable
Recreational	Tobacco; alcohol in large volume	Cessation; moderation

- Table answers the clinical question: **which substitution preserves the indication without the ED**; for antiandrogens no substitute exists.

Concept — A drug's sexual effect is predictable from its receptor profile: anything that blocks dopamine, raises serotonin or prolactin, removes androgen, or adds sympathetic/α₂ tone harms function, whereas agents that block RAS or potentiate NO spare or help it.

Toxic Exposures

- Environmental toxins act by **modulating androgens, systemic inflammation, oxidative stress and vascular disruption** → ED a plausible sequela.

- Clearly linked to impaired penile haemodynamics in rodents: **bisphenol A, gasoline exhaust and other atmospheric pollutants, pesticides, phthalates, heavy metals.**

Age

- **Healthy ageing:** ↑ latency to erection, ↓ turgidity, ↓ ejaculatory force and volume, longer refractory period, ↓ penile sensitivity, ↓ nocturnal erections.
- Cause incompletely understood — tissue and circulatory changes of age-associated conditions and **age-related androgen decline**. Animal ageing: ↓ smooth muscle-to-collagen, ↓ nNOS and eNOS, ↑ apoptosis and oxidative stress, ↑ RhoA/Rho-kinase, ↑ RAS, ↓ **angiotensin-(1-7)**, upregulated long noncoding RNAs.

Metabolic diseases

- Chronic conditions share **inflammation and oxidative stress**. Inflammatory markers linked to ED in DM and obesity: **C-reactive protein (CRP), toll-like receptors, interleukin (IL)-1, IL-6, IL-16, chemokines CCL2, CXCL1, CXCL8, TNF-α.**

Diabetes mellitus

- **ED much more prevalent in DM and presents at younger ages;** also common in **prediabetes.**
- Mechanisms: **penile arterial disease, ↓ androgen, peripheral neuropathy, impaired endothelium-dependent relaxation, loss of smooth muscle structure and function, endothelial apoptosis, oxidative stress.**
- Animal models: endothelial dysfunction with **incompetent cavernous endothelial cell-cell junctions;** ↓ nNOS, ↓ eNOS activity, oxidative stress, ↑ **advanced glycation end products (AGEs)**, ↓ elastin, ↓ VEGF, hypercontractility, ↓ **smooth muscle-to-collagen → veno-occlusive failure;** ↓ **AR expression via microRNA-205.**
- Functional mediators:
 - ↓ **endothelial small- ($K_{Ca}2.3$) and intermediate-conductance ($K_{Ca}3.1$) K^+ channels → ↓ ACh relaxation (db/db mice).**
 - **Ras / mitogen-activated protein kinase / peroxisome proliferator-activated receptor pathway and mitochondrial 3-hydroxy-3-methylglutaryl-CoA synthase 2 activation (inhibition mitigates).**
 - ↓ **Myosin phosphatase target subunit 1 (MYPT1) → ↓ MLCP → impaired relaxation.**
 - **Wnt10b upregulation → fibrosis.**
 - **Angiotensin II → TGF-β and RhoA/Rho-kinase → structural change and oxidative DNA damage; captopril, losartan or angiotensin-(1-7) ameliorate.**
 - **Rho-kinase → ↑ vascular arginase → impaired relaxation.**
 - ↑ **Insulin-like growth factor binding protein 3 (IGFBP3);** its inhibition improves erection.
 - **Neuropathy from AGE effects on neural microcirculation; impaired H_2S metabolism.**

TABLES 71.11 AND 71.12 — SUMMARY FINDINGS IN DIABETIC MEN AND DIABETIC ANIMAL MODELS (SECOND INTERNATIONAL CONSULTATION ON SEXUAL MEDICINE; SAENZ DE TEJADA ET AL., 2005)

Domain	Men with diabetes (Table 71.11)	Animal models (Table 71.12)
Anatomy / vessels	Atheroma in large vessels; pudendal and iliac stenosis; high prevalence of penile arterial insufficiency on Doppler	—
Erectile function	↓ Frequency and rigidity of nocturnal erections; ↓ rigidity after intracavernous vasodilator	—

Domain	Men with diabetes (Table 71.11)	Animal models (Table 71.12)
Ultrastructure	↓ Smooth muscle, ↑ collagen, thickened basal lamina , endothelial loss; ↑ smooth muscle apoptosis	—
NO pathway	↓ eNOS- and nNOS-mediated relaxation with preserved NO-donor relaxation ; ↓ insulin-mediated L-arginine and NADPH transport; ↑ arginase II depleting L-arginine	Streptozotocin rodents: selective nitrergic nerve degeneration ; ↓ endothelial and neurogenic NO relaxation; ↓ L-arginine; glycated haemoglobin impairs endothelium-dependent relaxation
EDHF	↓ EDHF-mediated relaxation in penile resistance arteries	Impaired EDHF responses
Contractile side	↑ α-adrenergic contraction in type 1, not type 2	↑ Tone via upregulated endothelin-A receptors ; ↑ contractile PGs and free radicals; ↑ prostacyclin synthesis
Other	↑ AGEs; ↑ TNF-α	Diabetic rabbit: ↓ cAMP response to PGE₁ / forskolin after 6 months ; ↑ glucose-induced protein kinase C via oxidative stress

- Pattern: **diabetes damages the NO source (nerve and endothelium) while sparing the smooth muscle's response to exogenous NO** — the basis for NO-donor preservation and PDE5I use, until structural fibrosis supervenes.

Metabolic syndrome

- **MetS = glucose intolerance, insulin resistance, dyslipidaemia, HTN**, often with visceral adiposity; **strongly linked to ED**; associated with ageing and ↓ T.
- ↑ **Oestrogen** in MetS; an **oestrogen receptor antagonist mitigated** ↓ smooth muscle and impaired ACh relaxation in an animal model.
- Endothelial dysfunction, inflammation (↑ TNF-α, CRP, IL-6), adipokine **resistin**; **nNOS uncoupling** and dysfunctional nitrergic relaxation (Fig. 71.13). **Exercise reversed** cavernous changes in high-fat-fed rabbits.

Renal insufficiency/failure

- ↓ Desire, ED and ejaculatory impairment common. **Up to 80% of men with end-stage renal disease have arterial and veno-occlusive ED.**
- Uraemic contributors: **hypothalamic-pituitary-testicular disturbance, hyperprolactinaemia, accelerated atheroma**; ↓ NO production/bioavailability, **pudendal artery calcification**, ↓ endothelial reactivity; **autonomic neuropathy** (abnormal vascular and bulbocavernosus reflexes) — nerves chronically depolarised, likely **potassium-related, mitigated by dialysis**; **depressive symptoms independently predict** dysfunction on haemodialysis.
- **Transplantation may improve function, but ED persists in about half.**

Hyperuricaemia

- ↑ Uric acid → worse erectile function; animals: ↓ NOS isoforms, oxidative stress.

COVID-19

- ED risk factors overlap with predictors of severe **severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)** illness; **cytokine storm, hyperinflammation, vascular injury** may ↑ ED. Spike protein found in **corporal endothelium** (case report). **Durability and mechanism unclear.**

Other chronic medical conditions

- **HIV appears a direct cause**, plus stress (transmission fear, body image, comorbidity, stigma, obligatory condom use) → psychogenic ED.

- Also: cirrhosis, inflammatory bowel disease, heart failure, chronic obstructive pulmonary disease, **obstructive sleep apnoea, BPH/lower urinary tract symptoms (LUTS)** — via illness-related stress and direct systemic effects.

Nutrient deficiencies

- **Hyperhomocysteinaemia** — vascular disease and ED; from deficiency of **vitamins B6, B12, folate** and genetic defects such as **5,10-methylenetetrahydrofolate reductase (MTHFR)** polymorphisms.
- **Folate**: inverse association with ED, possibly independent of homocysteine — **conflicting data**. **Low vitamin D** ↑ ED risk. Mechanisms likely vascular.

Concept — *Almost every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF- β -driven fibrosis — which is why ED behaves as a sentinel for cardiometabolic disease.*

Primary Erectile Dysfunction

- **Lifelong inability to initiate and/or maintain erection from the first sexual encounter or masturbation.**
- **Primary psychological ED**: performance anxiety and/or abnormal desire from **adverse childhood events, traumatic early sexual experience, conception concerns, misinformation**. All primary ED likely has a psychological dimension.
- **Congenital organic causes**: hypoplastic cavernous arteries; aberrant cavernous venous drainage (venous leak); congenital isolation of the corporal bodies; fusiform megalourethra with distal corporal atresia; congenital absence of the corpora.
- **Micropenis**: functionally normal but small — **stretched length more than 2.5 standard deviations below the mean for age** — may present as "ED".
- **Functional ED with normal haemodynamics** — **penile buckling**: axial load tolerance depends on intracavernous pressure, penile geometry and corporal expandability; consider **curvature (chordee, Peyronie disease)** when rigidity is full but intercourse fails.

KEY POINTS: OTHER CAUSES OF ERECTILE DYSFUNCTION

- **Ageing** contributes through central regulation, hormones, neuronal integrity, vascular flow and penile structure.
- **DM, MetS and cavernous nerve injury** cause particularly **severe** ED phenotypes.
- ED accompanies many chronic conditions and **may be a sentinel event for serious disease**.
- Drugs most commonly implicated: **antiandrogens, antidepressants, antihypertensives**.
- **Primary ED**: psychogenic causes, congenital penile anomalies or vascular defects.

Concept — *Lifelong ED is psychogenic until proven otherwise but not exclusively so; the organic causes are developmental failures of the same three components — arteries, veins and corporal bodies — that fail later in acquired ED.*

PERSPECTIVES

- Progress in erectile physiology has **revolutionised care** of ED; molecular biology, signal transduction, growth factors, microarrays and **stem cells** will continue to improve it.
- **The psychosocial context of human sexuality remains essential**, whatever the basic-science advances.